Elaborated-Structure Awareness SAR Imagery Using Hessian-Enhanced TV Regularization

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***Abstract*— Due to the sparse feature enhancement only concentrating on strong scatterers of target of interest, the conventional sparsity-driven synthetic aperture radar (SAR) imagery often encounters the loss of elaborated-structure features, where weak scatterers would be overlapped by the sidelobes of strong scatterers. In this article, an elaborated-structure awareness SAR (ESA-SAR) imaging algorithm is proposed based on Hessian-enhanced total variation (HETV) regularization and cooperation. By encoding the Hessian operator onto the prior of the interested target, the high-order information connected with elaborated-structure features of interests can be captured. Different from the conventional high-order formulation that is projected onto Euclidean norm balls, the proposed algorithm uses the Schatten norm balls as the projection space, where the high-order structure tensor is established, and the elaborated-structure feature can be extracted under the intended convex regularizer. More specifically, the HETV regularizer is analytically solved under the proximal algorithm considering its nondifferentiability. An eigen-soft-thresholding (E-ST) operator is derived, so that a closed-form solution for the elaborated-structure feature can be obtained. Moreover, a synergistic multitask learning framework embedded with the sparse feature enhancement is introduced, in which the elaborated-structure feature can be solved in a cooperative manner. The cooperative learning is guaranteed in terms of both theoretical and practical aspects. Finally, both simulated and raw SAR data are processed to validate the effectiveness of the ESA-SAR algorithm. Comparisons with conventional algorithms examine the superiority of the proposed algorithm.**

***Index Terms*— Hessian matrix, proximal algorithm, Schatten norm, synthetic aperture radar (SAR), total variation (TV).**

I. Introduction

SYNTHETIC aperture radar (SAR) is a coherent microwave imaging system, which can work in all day-and-night, all weather, and long operational-range conditions. Due to the synthesis aperture generated by relative movements between the radar and target of interest, SAR is capable of generating high spatial resolution images [1], [2], [3], [4], [5]. The conventional SAR image formation processing (IFP) algorithms mainly focus on the correction of range migrations. Commonly, those algorithms are based on Fourier transformation [6], [7], which demands that the systematic sampling frequency satisfies the Nyquist theory. In such cases, the correlation between the spectrum and response is fixed, where the mainlobe width is limited by the spectral bandwidth and the sidelobes of strong scatterers will override the weak scatterers. The fixed response function will definitely limit the extraction of specific features. Therefore, the conventional SAR IFP algorithms based on Fourier transformation are incapable of quantitatively extracting key features of the interested target.

Compressive sensing (CS) paves a way to quantify and extract dominant features of target. It was first introduced to high-resolution radar imagery in 2007 by Baraniuk and Steeghs [8]. In CS, few measurements can be used to recover the sparse feature of target of interest [9], [10], which can not only effectively break through the limitation of Nyquist sampling rate but also achieve the super-resolution imagery. Greedy algorithm is a classical algorithm for sparsity reconstruction. Matching pursuit (MP) and orthogonal matching pursuit (OMP) are the typical algorithms in greedy manner [11], [12], which are inherently a least-square estimation. They execute in iterations under a proper sparsity constraint.

Alternatively, sparsity reconstruction can be implemented in a sparse feature characterization with analytical solutions, which includes Bayesian inference and convex regularization. Under Bayesian inference [13], [14], a statistical sparse prior can be encoded, and a closed-form solution for the posterior can be obtained in a hierarchical manner [15]. Laplace is a typical probability prior for sparsity reconstruction, in which the sparse prior can be encoded as a norm regularizer. Convex optimization is a solution framework [16], [17]. The ℓ_1 regularized minimization problem is known as the least absolute shrinkage and selection operator (LASSO), which can be considered as the optimal convex optimization sparse solution [18]. The accuracy and/or efficiency for the sparse feature recovery can be improved by various methods. However, both the Bayesian inference and convex regularization mainly emphasize on dominant scatterers of the target to be imaged, in which the dominant sparsity feature is quantitatively expressed in closed-form solutions. In such cases, weak scatterers containing elaborated features would be blocked by the enhanced sparse features. Nevertheless, the idea of quantitative characterization for the sparse feature is heuristic

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for the extraction of elaborated-structure which associates various detailed features.

To this end, multiple regularization frameworks are introduced in [19] and [20], which aims at enhancing the target while suppressing the background clutters. In [19], the regularization is established under a semi-quadratic model. In [20], the framework is developed by different semantic contents in a scene of interest, and the semantic causal rules are encoded as the high-level semantic prior knowledge. Furthermore, the total variation (TV) regularizer becomes a favorable candidate [21], [22]. Since the TV norm is capable of evaluating correlations of adjacent scatterers, it has been a popular regularizer in the feature enhancement. With linear operations such as gradients and divergences, TV is capable of powerfully recovering structural edges of target of interest. However, its optimization will result in artifacts and homogeneous regions for the same reason. To overcome the limitations of TV, several functional variants have been proposed. Introducing structure tensor into the variation, structure tensor TV (STV) emerges [23], [24], which uses eigenvalues and eigenvectors to improve the directional awareness for structures of target. However, STV only depends on first-order gradient, which is incapable of extracting elaborated-structure features with high-order variations. To this end, the total generalized variation (TGV) is introduced with its innovation that improves piecewise constant to piecewise linear by high-order formulation [25]. Nevertheless, TGV cannot take advantage of the structure information from eigenvalue and eigenvector. Consequently, it is difficult to achieve closed-form solutions for high-order features.

Followed by high-resolution SAR imagery, high frequency and large bandwidth bring possibilities to explore elaborated features. Although weak scatterers would be suppressed from the dominant scatterers when sparse feature is enhanced, it will be definitely beneficial to recover or enhance multiple key features simultaneously. Therefore, to obtain detailed features of target of interests, a multitask framework should be established, which is required to achieve a synergetic processing for multifeature enhancement. However, the conventional sparsity enhancement is either a large computational load or incapable to coordinate multiple features. Alternating direction method of multipliers (ADMM) is a simple but powerful tool in terms of multifeature enhancement [24], [26], which can realize the cooperative optimization for multiple regularizers under the variable splitting and data argumentation methods [27]. The variable splitting is capable of decomposing a joint optimization problem into several subproblems that would be easy to be solved [27]. While the data argumentation can coordinate the subsolutions and achieve a global optima [27]. Also, the convergence of ADMM can be guaranteed by the augmented Lagrangian. Due to the proximal algorithm embedded in the ADMM framework [28], closed-form solutions for the multiple features can be ensured, which will definitely facilitate the computations of the feature enhancement. Finally, regarding computational complexity, the proximity-based ADMM framework is superior in terms of nondifferential problems [26]. For the simulation data in the size of N and M in the number of pulses and range cells, respectively, the complexity of the ADMM algorithm can be evaluated in the order of $O(2M \log N)$, the conventional convex optimization (CVX) is in the order of $O(MN^2)$, and the conventional sparse Bayesian learning (SBL) is in the order of $O(MN \log N)$ [27]. Obviously, the ADMM framework is superior in terms of computational complexity.

In this article, to enhance elaborately-structural features of the scenes or targets, a novel elaborated-structure awareness SAR (ESA-SAR) imaging algorithm is proposed, which exploits the Hessian-enhanced TV (HETV) regularizer to characterize the high-order clustering features of the target. Specifically, the HETV regularizer projects the Hessian matrix onto the Schatten norm balls, where the high-order structure tensor is established and the elaborated-structure feature can be extracted under the fitting of variational directions. Furthermore, a closed-form solution of HETV regularizer is analytically derived under the proximal algorithm to facilitate the proposed algorithm. More specifically, the enhanced TV regularization is optimized under the ADMM framework and combined with the sparse prior, ℓ_1-norm, to achieve an accurate SAR imagery, where multiple features are collaboratively enhanced. It is worth noting that analytical solutions named proximal operator [28] of the HETV regularization and ℓ_1-norm are synergistically derived in the ADMM framework, which effectively avoids the error propagation between multiple tasks of feature enhancement. Finally, both the simulated and raw SAR data are applied for the validation of the proposed algorithm. Comparisons with the conventional algorithms are performed and evaluated in terms of both qualitative and quantitative aspects.

The organization of this article is as follows. A generalized airborne SAR signal model is given in Section II, and the HETV regularizer is introduced in Section III. In Section IV, the intended ESA-SAR imaging algorithm is explained in detail. In Section V, the processing flowchart of the ESA-SAR is shown. Both the simulation and raw SAR data experiments are presented in Section VI. Finally, conclusions are drawn in Section VII.

II. SAR Geometry and Signal Model

Fig. 1 shows a generalized imaging geometry in the Cartesian coordinates XYZ-O for airborne spotlight SAR, where the radar is assumed flying along a predetermined flight track, X-axis, with a constant velocity v and altitude H. By directing wideband pulses to the observing scene to be imaged, the echoed pulses are collected. Defining $\mathbf{q}_0$ as the reference position of the antenna phase center (APC) of the airborne SAR, it can be expressed by the slow time t_n as $\mathbf{q}_0 = vt_n\vec{\mathbf{x}}$, where $\vec{\mathbf{x}}$ is the unit vector of the azimuthal direction (along-track). In principle, the coherent phase information in receiving echoes is necessary for high-resolution SAR imagery [29], where the phase history is determined by the range variation between the moving radar and the target. Defining $\mathbf{R}_0$ as the nearest reference range vector from the observing scene center to the flight track, and assuming $\mathbf{r}^p$ is the offset of the pth stationary scatterer relative to the observing scene center, the range history of the stationary target for the reference APC

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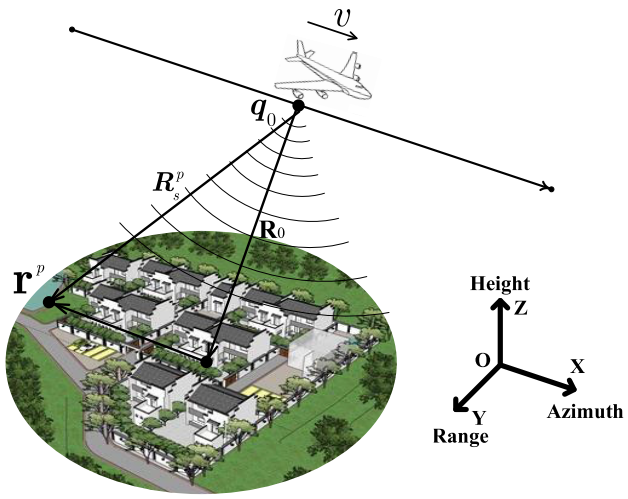


Fig. 1. Generalized airborne SAR geometry.

position can be written as

$$R_s^p(t_n) = \left|\mathbf{R}_0 + \mathbf{r}^p - \mathbf{q}_0(t_n)\right| \tag{1}$$

where $|\cdot|$ denotes the module operator of a vector. In (1), the range history is established by assuming that the target is stationary and the radar follows the predetermined flight track [30], [31], [32]. Next, by expanding the range history by the Taylor series up to the first order at $|\mathbf{R}_0 - \mathbf{q}_0(t_n)|$, there is

$$\begin{aligned} R_s^p(t_n) &= \left|\mathbf{R}_0 + \mathbf{r}^p - \mathbf{q}_0(t_n)\right| \\ &= \left|\mathbf{R}_0 - \mathbf{q}_0(t_n)\right| + \frac{\mathbf{R}_0 - \mathbf{q}_0(t_n)}{\left|\mathbf{R}_0 - \mathbf{q}_0(t_n)\right|}\mathbf{r}^p + O\left(\mathbf{r}^p\right) \end{aligned} \tag{2}$$

where $O(\mathbf{r}^p)$ includes high-order Taylor series.

Commonly, SAR transmits linear frequency-modulated pulses with bandwidth B and carrier frequency f_0. The echoed pulses are generally time-delayed signals of transmitting pulses with the summation of all the scatterers in the observing scene, which can be expressed as

$$S_0(k, t_n) = P(k)\sum_p \xi_p \exp\left[-\mathrm{j}k R_s^p(t_n)\right] + n_s(k, t_n) \tag{3}$$

where $k = 4\pi f/c$ denotes the wavenumber variable, $f = f_0 + \Delta f$ gives the transmitted frequency with the variation $\Delta f \in [-B/2, B/2]$, and c is the speed of light. In (3), the signal model in the spatial frequency domain is formulated, which consists of the transmitted pulse spectrum $P(k)$, and the backscattering intensity ξ_p of the pth stationary scatterers and the phase modulation from the systematic range history $R_s^p(t_n)$ are represented by exponential terms. Besides, $n_s(k, t_n)$ denotes the systematic noise. Defining $\vec{\mathbf{r}}$ to be the unit vector of slant range, which is orthogonal to the azimuth direction $\vec{\mathbf{x}}$, the following vectors can be expressed as $\mathbf{R}_0 = R_0\vec{\mathbf{r}}$ and $\mathbf{r}^p = x^p\vec{\mathbf{x}} + r^p\vec{\mathbf{r}}$, where x^p and r^p denote the azimuthal and slant-range positions of the pth stationary scatterer, respectively. Substituting the above vectors and the range equation in (2) into $S_0(k, t_n)$ in (3), the azimuth-dechirped signal [30] can be written as

$$S_0(k, t_n) = P(k)\sum_p \xi_p \exp\left[-\mathrm{j}k\left(\frac{R_0 r^p - v t_n x^p}{\sqrt{{R_0}^2 + \left(v t_n\right)^2}}\right)\right] + n_s(k, t_n). \tag{4}$$

It can be noted from (4) that the transmitting wavenumber k is coupled with slow time t_n, which results in the systematic range migration that will defocus the intended SAR image. Commonly, the polar format algorithm (PFA) can be used to remove the range migration. In general, it uses a 2-D resampler in the slant-range and azimuth directions for the decoupling or range migration correction [32], [33]. Mathematically, the resampler is to transform the data from the polar format (k, t_n) to a rectangle format (k_r, k_x), where $k_r = R_0 k/(R_0^2 + (vt_n)^2)^{1/2}$ and $k_x = -k_r v t_n/R_0$ are the resampled range wavenumber and azimuth wavenumber [30], [31], respectively. After PFA resampling, the echoed signal in (4) can be expressed as

$$S_0(k_r, k_x) = P(k_r)\sum_p \xi_p \exp\left[-\mathrm{j}\left(k_r r^p + k_x x^p\right)\right] + n_s(k_r, k_x). \tag{5}$$

In (5), the azimuth wavenumber k_x can be further expressed as $k_x = -k_0 v t_n/R_0$, where $\tilde{t}_n = -R_0 k_x/v k_0$ is linearly related with k_x, and it can be regarded as a resampled slow-time variable. In such cases, the range-compressed version of (5) can be given as

$$S_0\left(\hat{r}, \tilde{t}_n\right) = \sum_p \xi_p \mathrm{sinc}\left(\hat{r} - r^p\right)\exp\left(\mathrm{j}2\pi f_d^p \tilde{t}_n\right) + n_s\left(\hat{r}, \tilde{t}_n\right) \tag{6}$$

where $\hat{r}$ is the variable related to the slant-range position and f_d^p represents the Doppler frequency. In (6), $\mathrm{sinc}(\cdot)$ denotes the range response with respect to the range spectrum of $P(k_r)$ and the exponential term is related to the azimuthal position of the scatterer.

From the above discussion, the range-compressed signal model can be formulated in matrix form as

$$\mathbf{Y} = \mathbf{A}\mathbf{X} + \mathbf{N} \tag{7}$$

where $\mathbf{Y} \in \mathbb{C}^{N\times M}$ is the range-compressed data matrix denoting $S_0(\hat{r}, \tilde{t}_n)$ in (6), $\mathbf{N} \in \mathbb{C}^{N\times M}$ is the noise matrix, and N and M represent the number of echoed pulses and range cells, respectively. Furthermore, $\mathbf{A} \in \mathbb{C}^{N\times N}$ in (7) denotes the Fourier dictionary to form the intended SAR image $\mathbf{X} \in \mathbb{C}^{N\times M}$, which can be given as

$$\begin{aligned} \mathbf{A} &= \left[\mathbf{a}\left(f_d^1\right), \ldots, \mathbf{a}\left(f_d^N\right)\right] \\ \mathbf{a}\left(f_d^n\right) &= \left[e^{-j2\pi f_d^n \tilde{t}_1}, \ldots, e^{-j2\pi f_d^n \tilde{t}_N}\right]^{\mathrm{T}} \end{aligned} \tag{8}$$

where f_d^n indicates the Doppler frequency index, and $[\cdot]^{\mathrm{T}}$ gives the vector/matrix transpose operator. From (7), the SAR imagery is inherently a typical inverse problem. In noiseless circumstance, to solve the image $\mathbf{X}$, a simple matrix inversion of $\mathbf{A}$ can be applied for $\mathbf{X} = \mathbf{A}^{-1}\mathbf{Y}$. However, in noisy environment, especially, when the data are undersampled, the

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inverse problem will be ill-posed. Therefore, it is necessary to encode a proper prior knowledge for an accurate image/feature recovery.

III. HETV Regularizer

To overcome the ill-posed inverse problem in (7), there are generally two ways for the quantitative incorporation of prior knowledge. First, Bayesian inference is capable of encoding a prior in a statistical manner. However, the solvability for the posterior may be questionable when the prior to be incorporated is not a common one. In addition, the computational load of the Bayesian inference is large, which limits practical applications. Second, convex regularization is the other way to incorporate a quantitative prior. Generally, a monotropic norm function can be introduced as a regularizer and a convex optimization can be performed to achieve the estimation. It is worth noting that convex regularization is easy to handle and generalize, and it is easy to achieve closed-form solutions for the specific features to be enhanced.

A. Total Variation

In this article, considering a complicated prior is to be encoded for the elaborated feature to be enhanced, the convex regularization scheme is used, which can facilitate the incorporation of prior knowledge. First, advanced regularizers are to be investigated. To capture structural features, TV norm is the most common regularizer to be selected. It is capable of extracting structural features by evaluating variations among adjacent scatterers. Mathematically, the TV norm is formally defined as

$$\mathrm{TV}(\mathbf{X}) = \sup\left\{\int_{\mathbf{\Omega}} \mathbf{X}(\mathrm{div}\mathbf{V})dx \mid \mathbf{V} \in \mathbf{C}_c^1(\mathbf{\Omega}, \mathbf{C}^2), \mathbf{V}(x) \in B_2(0), \forall x \in \mathbf{\Omega}\right\} \quad (9)$$

where $\mathbf{X}$ denotes the image matrix, x is the resolution cell index of $\mathbf{X}$, div$\mathbf{V}$ is the divergence of the dual variable $\mathbf{V}$ that is relevant to the original variable $\mathbf{X}$, and the divergence operator performs as $\mathrm{div} : \mathbb{C}^{N\times M\times 2} \to \mathbb{C}^{N\times M}$. In (9), the dual variable $\mathbf{V}$ is defined in a joint space of $\mathbf{\Omega}$ and $\mathbf{C}^2$, where $\mathbf{C}^2$ denotes the tensor space. Furthermore, the element in $\mathbf{V}$ or $\mathbf{V}(x)$ is set in $B_2(0)$, which denotes a closed Euclidean unit ball centered at the origin. According to Appendix A, the adjoint relationship between the divergence operator in (9) and the gradient operator as $\nabla : \mathbb{C}^{N\times M} \to \mathbb{C}^{N\times M\times 2}$ is revealed, and the TV definition can be equivalently rewritten in the common discrete form as

$$\mathrm{TV}(\mathbf{X}) = \sum_x \|\nabla \mathbf{X}(x)\|_2 \quad (10)$$

where $\|\cdot\|_2$ is the ℓ_2-norm, and the detailed derivation from (9) to (10) is given in Appendix B. In (10), it is clear that the structure feature of $\mathbf{X}$ can be quantified by the TV norm with the summation of the gradients in the two fixed directions of range and azimuth, respectively. As a result, TV regularization is capable of reconstructing images with sharp, well-preserved edges. However, it has some shortcomings of oversmoothing homogeneous regions and creating staircase effects [23], which would limit the application of complicated target features.

B. HETV Regularizer

1) STV Regularizer: Conventionally, the structure tensor is introduced to improve the TV regularizer to accommodate for an arbitrary structural direction [34]. Therefore, STV regularizer can be generated as

$$\mathrm{STV}(\mathbf{X}) = \sum_x \|J\mathbf{X}(x)\|_{S_p} \quad (11)$$

where $J\mathbf{X}$ is the Jacobian operator of the image $\mathbf{X}$. In the 2-D SAR imagery, it has $J\mathbf{X} = \nabla\mathbf{X}$, where the Jacobian operator is inherently a first-order gradient operator. In (11), $\|\cdot\|_{S_p}$ denotes the pth-order Schatten norm (S_p-norm, $p \geq 1$). For the S_p-norm, setting $\mathbf{E} = J^{\mathrm{H}}\mathbf{X}(x)J\mathbf{X}(x)$ as the structure tensor of an arbitrary element in $\mathbf{X}$, the eigenvalue decomposition derives the eigenvalue as $\theta(\mathbf{E})$. Similarly, the singular value decomposition (SVD) for Jacobian matrix $J\mathbf{X}(x)$ derives as $J\mathbf{X}(x) = \mathbf{U}\Sigma\mathbf{V}^{\mathrm{H}}$, where $\mathbf{U}$ and $\mathbf{V}$ are consisted with the singular vectors, Σ is constructed by the singular values, and $(\cdot)^{\mathrm{H}}$ denotes the conjugate transpose operator. Then, the S_p-norm [23] of $\mathbf{E}$ can be further defined as

$$\|J\mathbf{X}(x)\|_{S_p} = \left(\sum_n^2 \sigma_n^p(J\mathbf{X}(x))\right)^{\frac{1}{p}} = \left(\sum_n^2 \theta_n^p(\mathbf{E})\right)^{\frac{1}{p}} \quad (12)$$

where σ_n and θ_n denote the nth singular value of $J\mathbf{X}(x)$ and eigenvalue of the structure tensor, respectively. From (11) to (12), the STV regularizer operates in the computation of the Schatten norm of Jacobian matrix of $\mathbf{X}$. It is worth noting that the STV regularizer generates the Schatten norm balls on which the Jacobian matrix can project on, and the structural information can therefore be extracted by the singular value analysis. Consequently, STV is capable of fitting clustering features of targets in associated directions. From the above-mentioned TV and STV regularizers, both of them are based on first-order gradient for the computation, in which only piecewise constant feature can be reconstructed. Therefore, both of them are difficult to formulate complicated features in high-order variations.

2) High-Order TV Regularizer: To accommodate high-order structural variations, a high-order version of TV regularizer named by TGV_α^2 is proposed in [35], which is capable of making a piecewise linear reconstruction. Specifically, TGV_α^2 can be given as

$$\mathrm{TGV}_\alpha^2(\mathbf{X}) = \sup\left\{\int_{\mathbf{\Omega}} \mathbf{X}\mathrm{div}^2\mathbf{V}dx \mid \mathbf{V} \in \mathbf{C}_c^2\left(\mathbf{\Omega}, \mathrm{Sym}^2\left(\mathbf{C}^2\right)\right), \mathbf{V}(x) \in \alpha_0 B_{2\times 2}(0), (\mathrm{div}\mathbf{V}(x))^{\mathrm{T}} \in \alpha_1 B_2(0), \forall x \in \mathbf{\Omega}\right\} \quad (13)$$

where $\mathrm{div}^2\mathbf{V}(x)$ is the second-order divergence for $\mathbf{V}(x)$, $\alpha_0 B_{2\times 2}(0)$ denotes a Euclidean norm ball, $\mathrm{Sym}^2(\mathbf{C}^2)$ denotes a symmetric tensor space, and $\mathbf{V}(x)$ is a 2×2 symmetric tensor of dual variable $\mathbf{V}$. In (13), it can be noted that the piecewise linear reconstruction of TGV regularizer is realized

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by high-order divergence and the Euclidean norm is the high-order quantification formulation.

3) HETV Regularizer: In this article, a novel TV regularizer is designed to capture elaborated direction and high-order variation in structures of the interested target. It takes advantage of the high-order gradient/divergence in the Hessian matrix and the associated Schatten norm balls on which elaborated variations can be formulated. Mathematically, the HETV regularizer is defined as

$$\begin{aligned}\mathrm{HETV}(\mathbf{X}) = \sup\Bigg\{ \int_{\mathbf{\Omega}} \mathbf{X}\mathrm{div}^2\mathbf{V}dx \mid \mathbf{V} \in \mathbf{C}_c^2(\mathbf{\Omega}, \mathbf{C}^2), \\ \mathbf{V}(x) \in \alpha_0 B_{S_q \times S_q}(0), \\ (\mathrm{div}\mathbf{V}(x))^{\mathrm{T}} \in \alpha_1 B_{S_q}(0), \forall x \in \mathbf{\Omega} \Bigg\} \end{aligned} \tag{14}$$

where $\alpha_0 B_{S_q \times S_q}(0)$ denotes a S_q-norm ball and S_q-norm is the dual norm of S_p-norm under the parameter relationship as $(1/q) + (1/p) = 1$. In (14), the second-order variation in neighboring scatterers can be captured for piecewise linear recovery of structure features because of second-order divergence computation, which is similar to the TGV regularizer shown in (13). However, different from the Euclidean norm projection in (13), the intended regularizer projects the high-order divergence onto the Schatten norm balls, on which the structure tensor can be used for high-order eigen information. Therefore, the structural details can be evaluated by the directional eigen information to effectively capture complicated structural features.

Based on the adjoint relationship between divergence and gradient operator [23], the second order of divergence $\mathrm{div}^2\mathbf{V}(x)$ can be equivalently expressed under the second-order difference as

$$\mathrm{div}^2\mathbf{V}(x) = \Delta^2_{x_1x_1}(V_{11}) + \Delta^2_{x_2x_2}(V_{22}) + \left(\Delta^2_{x_1x_2} + \Delta^2_{x_2x_1}\right)(V_{12}) \tag{15}$$

where $\Delta^2_{x_1x_2} = \Delta^2_{x_2x_1}$, and $\Delta^2_{x_1x_1}$ and $\Delta^2_{x_2x_2}$ denote the backward finite-difference operators with Neumann boundary conditions. Next, tensor $\mathbf{V}(x)$ is symmetric, which can be given as

$$\mathbf{V}(x) = \begin{bmatrix} V_{11}(x) & V_{12}(x) \\ V_{21}(x) & V_{22}(x) \end{bmatrix}. \tag{16}$$

According to (15) and (16), the intended regularizer $\mathrm{HETV}(\mathbf{X})$ in (14) can be rewritten in the gradient form as

$$\mathrm{HETV}(\mathbf{X}) = \sum_x \|H\mathbf{X}(x)\|_{S_p} \tag{17}$$

where the Hessian operator $H\mathbf{X} : \mathbb{C}^{N\times M} \to \mathbb{C}^{N\times M\times 2\times 2}$ denotes the second-order derivative with respect to $\mathbf{X}$. Details about the derivation of (17) are provided in Appendix B. In (17), $H\mathbf{X}(x) \in \mathbb{C}^{2\times 2}$ is the second-order structure tensor, which can be given in a discrete form as

$$H\mathbf{X}(x) = \begin{bmatrix} \nabla^2_{x_1x_1}(x) & \nabla^2_{x_1x_2}(x) \\ \nabla^2_{x_2x_1}(x) & \nabla^2_{x_2x_2}(x) \end{bmatrix} \tag{18}$$

where $\nabla^2_{x_1x_1}$ and $\nabla^2_{x_2x_2}$ denote the forward finite-difference operators that approximate the second-order partial derivatives along the x_1 and x_2 dimensions, respectively, and $\nabla^2_{x_1x_2} = \nabla^2_{x_2x_1}$ denotes the second-order mixed partial derivatives.

Next, SVD can be used for structure tensor decomposition. However, the computational load is too large for practices. Considering the symmetry of the Hessian matrix in (18), the eigen decomposition can be used to equivalently replace SVD computation. Denoting $\lambda^+_H(x)$ and $\lambda^-_H(x)$ to be the eigenvalues of the Hessian matrix, where $\lambda^+_H(x) \geq \lambda^-_H(x) \geq 0$, the intended regularizer in (17) can be accordingly derived as

$$\mathrm{HETV}(\mathbf{X}) = \sum_x \left\| \left(\sqrt{\lambda^+_H(x)}, \sqrt{\lambda^-_H(x)} \right) \right\|_p \tag{19}$$

where $\|\cdot\|_p$ is the ℓ_p-norm of vectors. It can be noted from the derivations in (17)–(19) that the intended HETV regularizer is originally established as a S_p-norm of the Hessian matrix. Considering the summation across x, the HETV regularizer can be simply expressed by vector norm $\|\cdot\|_{1,p}$, which is the mixed version of ℓ_1 and S_p-norm. Therefore, the HETV regularizer in (19) can be explicitly denoted as

$$\mathrm{HETV}(\mathbf{X}) = \|H\mathbf{X}\|_{1,p} \tag{20}$$

where the vector norm $\|\cdot\|_{1,p}$ is defined as $\|\mathbf{X}\|_{1,p} = \sum_{n=1}^{N\times M} \|\mathbf{X}_n\|_{S_p}$. In (20), the intended HETV regularizer is capable of capturing both the high-order and directional variations in the structural features of the target of interests. First, the Hessian operator shown in (18) extracts high-order variations in the structural features to achieve piecewise linear reconstructions, which can better approximate the intensity variations in scatterers than the piecewise constant reconstructions provided by the TV/STV regularizers. Second, the projection of Schatten norm is further applied to obtain elaborately directional variation features. Therefore, the eigen decomposition is used for the analysis of high-order Hessian computation, and the intended HETV regularizer is finally represented as norm $\|\cdot\|_{1,p}$ of the Hessian matrix.

To visually demonstrate the mechanism of the intended HETV regularizer, Fig. 2 is shown. Meanwhile, the conventional first-order STV and second-order TGV^2_α regularizers are intuitively given to demonstrate the superiority of the HETV regularizer. In Fig. 2, the concentric circles are simulated for the structures to be imaged, which includes complicated multilevel structure features. The denser the contour, the larger the gradient of the features, such as point A in Fig. 2, and vice versa for point B. The first-order structure tensor is visualized as an ellipse with its eigenvectors θ^+_J and θ^-_J and the eigenvalues λ^+_J and λ^-_J, where $(\lambda^+_J(x))^{1/2}\theta^+_J$ depicts the direction of the gradient.

Based on the extraction of first-order gradient, the scatterers in the simulated contours can be mapped into the gradient curve as the curve line at the bottom of Fig. 2. In other words, after gradient computation, the structural features of the simulated target can be transformed into the gradient curve, where the denser the simulated contours, the larger the gradient value, and vice versa. Furthermore, as derived in (20), the intended Hessian-enhanced TV operates with the second-order structure tensor, which computes the Hessian matrix eigenvectors θ^+_H and θ^-_H and eigenvalues λ^+_H and λ^-_H. It is worth noting that the eigen information provided by the second-order tensor is capable of fitting the structural features of the targets by the Schatten norm projection. At point A,

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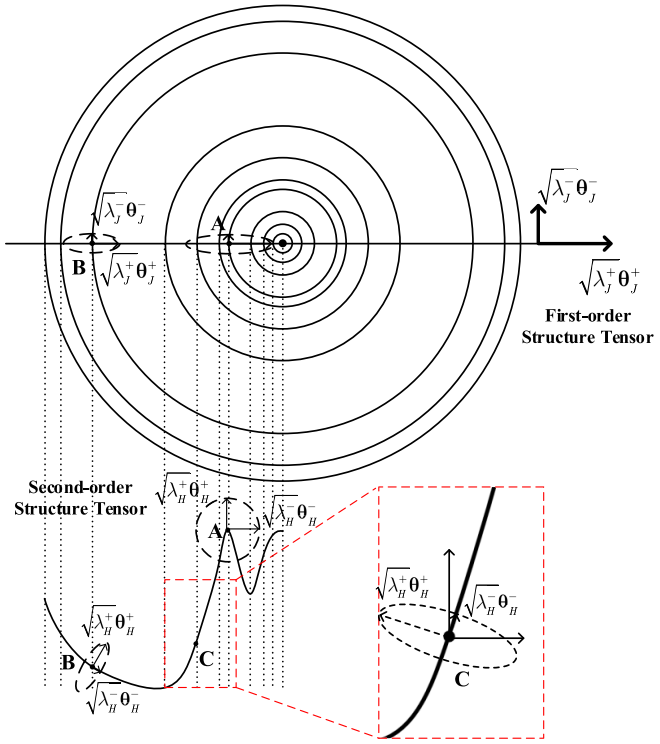


Fig. 2. Schematic of HETV regularizer.

the gradient value at the mapped curve line is the maxima, which is the inflection point, where the second-order tensor value equals to zero. This means the fast variation in the structural feature. On the contrary, at point B, the gradient value at the mapped curve line reaches to minima, which is also the inflection point of the gradient curve. However, the second-order tensor value determines the minima. In such a case, the slow variation in structural features can be captured. More generally, at point C, the gradient value corresponds to the structural feature variation, and the second-order tensor determines the direction of variation as the enlarged area shown in Fig. 2, where the direction is determined by the Hessian matrix eigenvalue and eigenvector different from that by the conventional $\mathrm{TGV}_{\alpha}^{2}$. It is obvious that the second-order tensor is capable of accurately representing the tangent and normal directions of the gradient direction. Consequently, the intended HETV regularizer can adaptively fit the high-order features. Moreover, the HETV regularizer projects the high-order variation onto the Schatten norm balls, instead of the conventional $\mathrm{TGV}_{\alpha}^{2}$ that onto the Euclidean norm balls. It inherently facilitates the formulation of elaborated-structure feature.

In this article, the intended HETV regularizer in (20) is used, and the regularization problem of elaborated-structure feature enhancement can be expressed as

$$\min_{\mathbf{X}} \ (1/2) \ \|\mathbf{AX}-\mathbf{Y}\|_F^2 + \xi \ \mathrm{HETV}(\mathbf{X}) \tag{21}$$

where ξ denotes the regularization parameter that can balance the function of regularization, and $\|\cdot\|_F$ is the Frobenius norm that measures the recovery accuracy. The problem in (21) is inherently an inverse problem, and the solution will be detailedly discussed in the following.

IV. ESA-SAR Imaging Algorithm

In this article, the intended HETV regularizer will be incorporated as a prior to enhance the elaborated-structure feature of the target of interests. Specifically, to guarantee the performance of feature enhancement under the conditions of low signal-to-noise ratio (SNR) and undersampling rate (USR), the sparse prior is also encoded [36]. Furthermore, both the priors are encoded and solved in a cooperative manner, so that the features can be highlighted collaboratively. To this end, the HETV and ℓ_1-norm are formulated and evaluated in a joint regularization as

$$\min_{\mathbf{X}} \ (1/2) \ \|\mathbf{AX}-\mathbf{Y}\|_F^2 + \xi_1 \ \mathrm{HETV}(\mathbf{X}) + \xi_2 \|\mathrm{vec}(\mathbf{X})\|_1 \tag{22}$$

where ξ_1 and ξ_2 are the regularization parameters corresponding to the priors of HETV and ℓ_1-norm, respectively, which determines the weight of priors. In (22), $\mathrm{vec}(\cdot)$ denotes the vectorized operator of a matrix.

To solve the joint optimization problem in (22), the ADMM framework is used in this article. The collaborative learning of the proposed algorithm is guaranteed by the "decomposition–coordination" framework in terms of the following five aspects. In terms of theory, the ADMM framework combines the splitted variable to decompose the multifeature joint enhancement problem into multiple subproblems, and then combines the data augmentation to coordinate the analytical solutions of multiple features to obtain a global optimization [27]. In terms of mechanism, the framework uses the idea of "decomposition–coordination" which decomposes the global optimization into multiple subproblems for the features' enhancement to effectively obtain the closed-form solutions of multiple features, which is finally coordinated to optimize the global optimization under the coordinate descent method. In terms of process, the framework follows the "decomposition–coordination" idea to achieve local optimization of features' enhancement and global cooperative optimization, where each step is updated iteratively, and the feature variables and global variable interact with each other. In terms of operation, the enhancement of different features is independent and does not interfere with each other, and the global optimal solution will coordinate the solutions of multiple features. In terms of accuracy, the framework introduces dual variables as residual variables to control the iterative accuracy, which can balance the enhancement of multiple features. Furthermore, the designated framework commonly uses proximal algorithm for closed-form solutions of multiple features, which greatly decreases computational complexity.

Since there are elaborated-structure and sparse features to be enhanced in this article, two splitted variables are to be introduced, and there will be two local problems to be optimized in a distributed manner. In such cases, the joint problem in (22) can be formulated in the ADMM manner as

$$\min_{\mathbf{X},\mathbf{Z}} \ f(\mathbf{X}) + g(\mathbf{Z}), \quad \text{s.t. } \mathbf{KX}-\mathbf{PZ}=\mathbf{S} \tag{23}$$

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where $f(\mathbf{X}) = (1/2)\|\mathbf{AX} - \mathbf{Y}\|_F^2$ is the data fidelity measuring the accuracy of the feature recovery, and $\mathbf{Z} = [\mathbf{Z}_1, \mathbf{Z}_2]^{\mathrm{T}}$ is a set of splitted variables from the primal variable $\mathbf{X}$, which corresponds to the intended features of target of interest. It is worth noting that the mapping relationship between the splitted variables and primal variable is decided in the equality of (23). In (23), $g(\mathbf{Z}) = \sum_I \xi_i g_i(\mathbf{Z}_i)$ represents a joint regularizer function including multiple regularizers for feature enhancement, and $\mathbf{K}$, $\mathbf{P}$ and $\mathbf{S}$ are the constant coefficient matrices, which constrain the equivalent relationship between $\mathbf{X}$ and $\mathbf{Z}$. Next, according to the ADMM procedure, the constrained problem in (23) can be transformed by augmented Lagrange and rewritten in the unconstrained form as

$$L(\mathbf{X}, \mathbf{Z}; \mathbf{U}) = f(\mathbf{X}) + g(\mathbf{Z}) + \mathbf{U}^{\mathrm{H}}(\mathbf{KX} - \mathbf{PZ} - \mathbf{S}) + \frac{\rho}{2}\|\mathbf{KX} - \mathbf{PZ} - \mathbf{S}\|_2^2 \quad (24)$$

where $(\rho/2)\|\mathbf{KX} - \mathbf{PZ} - \mathbf{S}\|_2^2$ is the augmented Lagrangian term, $\mathbf{U}$ is the dual variable corresponding to the splitted variable, and ρ is the augmented Lagrangian parameter, which determines the weight of the augmented Lagrangian term. The dual variable measures the residual between the primal variable $\mathbf{X}$ and the splitted variable $\mathbf{Z}$, which will be exploited to evaluate the accuracy for global optimization. Submitting $\mathbf{K} = [\mathbf{K_1}, \mathbf{K_2}]^{\mathrm{T}}$, $\mathbf{P} = \mathbf{I}$, $\mathbf{S} = \mathbf{0}$, the solution for (24) can be carried out in the coordinate descent manner as

$$\begin{cases} \text{for } i = 1:2 \\ \quad \mathbf{Z}_i^{k+1} = \arg\min_{\mathbf{Z}_i}\left\{\xi_i g_i(\mathbf{Z}_i) + \frac{\rho}{2}\left\|\mathbf{K}_i\mathbf{X}^k - \mathbf{Z}_i + \mathbf{D}_i^k\right\|_F^2\right\} \\ \quad \mathbf{D}_i^{k+1} = \mathbf{D}_i^k + \mathbf{K}_i\mathbf{X}^k - \mathbf{Z}_i^{k+1} \\ \text{end} \\ \mathbf{Z}^{k+1} = \left[\mathbf{Z}_1^{k+1}, \mathbf{Z}_2^{k+1}\right] \\ \mathbf{D}^{k+1} = \left[\mathbf{D}_1^{k+1}, \mathbf{D}_2^{k+1}\right] \\ \mathbf{X}^{k+1} = \arg\min_{\mathbf{X}}\left\{f(\mathbf{X}) + \frac{\rho}{2}\left\|\mathbf{KX} - \mathbf{Z}^{k+1} + \mathbf{D}^{k+1}\right\|_F^2\right\} \end{cases} \quad (25)$$

where the superscript k represents the number of iterations, and $\mathbf{D} = \mathbf{U}/\rho$ is the scaled dual variable. It can be noted from (25) that the procedure proceeds in a "decomposition–coordination" manner. The inner "for" iteration executes decomposed optimizations for splitted variables and their corresponding dual variables. Then, all the splitted variables and dual variables are collected and gathered into $\mathbf{Z}^{k+1}$ and $\mathbf{D}^{k+1}$. Finally, the global variable $\mathbf{X}^{k+1}$ is accordingly updated under the splitted variable $\mathbf{Z}^{k+1}$ and dual variable $\mathbf{D}^{k+1}$. Furthermore, to determine the specific values of the parameters ξ_i and ρ, maximum likelihood estimation is used, where the likelihood function can be established by iteratively generating the data likelihood [27]. Then, Newton iteration can be applied to achieve the scalar value of the parameters.

A. Elaborated-Structure Feature Enhancement

According to (25), first, the elaborated-structure feature will be evaluated, where the HETV regularization is optimized, and the elaborated-structure feature can be enhanced. Under the definition of the HETV regularizer in (20), the first splitted variable $\mathbf{Z}_1$ is introduced. By setting coefficient matrix $\mathbf{K}_1$ as the Hessian operator, the first splitted variable or the elaborated-structure feature variable $\mathbf{Z}_1$ can be given as $\mathbf{Z}_1 = H\mathbf{X}$. It should be noted that the Hessian operator $H(\cdot)$ corresponds to the coefficient matrix $\mathbf{K}_1$ and the splitted variable $\mathbf{Z}_1 \in \mathbb{C}^{N\times M\times 2\times 2}$ has a different dimension from the primal variable $\mathbf{X}$. The dimensionality will be decreased in the following coordination stage. By substituting the HETV definition in (20), the elaborated-structure feature variable can be solved as

$$\mathbf{Z}_1^{k+1} = \arg\min_{\mathbf{Z}_1} \xi_1\|\mathbf{Z}_1\|_{1,p} + \frac{\rho_1}{2}\left\|H\mathbf{X}^k - \mathbf{Z}_1 + \mathbf{D}_1^k\right\|_\chi^2 \quad (26)$$

where ρ_1 denotes the elaborated-feature-related Lagrangian parameter. The norm $\|\cdot\|_\chi$ corresponding to the high-order space $\mathbb{C}^{N\times M\times 2\times 2}$ is defined as

$$\|\mathbf{X}\|_\chi = \left(\sum_{n=1}^{N\times M}\|\mathbf{X}_n\|_F^2\right)^{\frac{1}{2}}. \quad (27)$$

It can be noted from (25) that the splitted variable $\mathbf{Z}_1$ and the dual variable $\mathbf{D}_1$ are defined in high dimensions $\mathbb{C}^{N\times M\times 2\times 2}$, which represent the high-order variations in the elaborated feature.

Due to the separability of $\|\cdot\|_{1,p}$ and $\|\cdot\|_\chi$ norms, the high-dimensional optimization in (26) can be further decomposed in $N \times M$ independent suboptimizations as

$$\left(\mathbf{Z}_1^{k+1}\right)_n = \arg\min_{(\mathbf{Z}_1)_n} \xi_1\|(\mathbf{Z}_1)_n\|_{S_p} + \frac{\rho_1}{2}\left\|\left(H\mathbf{X}^k - \mathbf{Z}_1 + \mathbf{D}_1^k\right)_n\right\|_F^2. \quad (28)$$

Under the substitution of SVD as $\mathbf{E} = \mathbf{U}\Sigma\mathbf{V}^{\mathbf{H}}$, and setting $\mathbf{E}_n = (H\mathbf{X}^k + \mathbf{D}_1^k)_n$, the proximity operator of the independent feature variable $(\mathbf{Z}_1)_n$ can be solved as

$$\begin{aligned}\left(\mathbf{Z}_1^{k+1}\right)_n &= \mathrm{prox}_{\frac{\xi_1}{\rho_1}\|\cdot\|_{S_p}}\left[\left(H\mathbf{X}^k + \mathbf{D}_1^k\right)_n\right] \\ &= \mathbf{U}_n\mathrm{diag}\left[\mathrm{prox}_{\frac{\xi_1}{\rho_1}\|\cdot\|_p}(\boldsymbol{\sigma}_n)\right]\mathbf{V}_n^H \end{aligned} \quad (29)$$

where prox$(\cdot)$ is the proximal operator, diag$(\cdot)$ is the transformation operator from a vector to a diagonal matrix, and $\boldsymbol{\sigma}_n$ is a vector whose elements are the singular values of $\mathbf{E}_n$. Setting $p = 1$, the problem in (29) can be derived as an eigen-soft-thresholding (E-ST), which is denoted as

$$\begin{aligned}\left(\mathbf{Z}_1^{k+1}\right)_n &= \mathbf{U}_n\mathrm{diag}\left[\mathrm{prox}_{\frac{\xi_1}{\rho_1}\|\cdot\|_1}(\boldsymbol{\sigma}_n)\right]\mathbf{V}_n^H \\ &= \mathbf{U}_n\mathrm{diag}\left[\left[\boldsymbol{\sigma}_{\mathbf{n}} - \frac{\frac{\xi_2}{\rho_2}\boldsymbol{\sigma}_{\mathbf{n}}}{|\boldsymbol{\sigma}_{\mathbf{n}}|}\right]_+\right]\mathbf{V}_n^H \end{aligned} \quad (30)$$

where $[\cdot]_+$ denotes the soft-thresholding operator [27] to obtain the sparse feature. It is worth noting that the soft-thresholding operator is the proximal operator of ℓ_1-norm due to the nondifferentiability of ℓ_1-norm. The soft-thresholding operator is a closed-form solution with piecewise linear property. For the sake of convenience, the update function for the feature variable $\mathbf{Z}_1$ at the $(k+1)$th iteration can be written as

$$\mathbf{Z}_1^{k+1} = \mathrm{prox}_{\frac{\xi_1}{\rho_1}\|\cdot\|_{1,p}}\left[H\mathbf{X}^k + \mathbf{D}_1^k\right] \quad (31)$$

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where $\text{prox}_{(\xi_1/\rho_1)\|\cdot\|_{1,p}}$ is the proximal operator of $\|\cdot\|_{1,p}$-norm, which denotes the solution of elaborated-structure feature. In (31), the solution for the elaborated-structure feature is obtained as a closed-form function of E-ST operator. It is linear, and the elaborated-structure feature enhancement is phase history guaranteed. Obviously, the Hessian transformation $H\mathbf{X}$ for the image matrix $\mathbf{X}$ should be determined previously, and the dual variable $\mathbf{D}_1$ at the kth iteration is also required. To evaluate the accuracy of the solution of the splitted variable $\mathbf{Z}_1$, the corresponding dual variable $\mathbf{D}_1$ is derived and updated as

$$\mathbf{D}_1^{k+1} = \mathbf{D}_1^k + H\mathbf{X}^k - \mathbf{Z}_1^{k+1}. \tag{32}$$

In (31) and (32), the solutions for elaborated-structure variable $\mathbf{Z}_1$ and dual variable $\mathbf{D}_1$ are obtained in closed-form, respectively. It can be regarded as the local solution for the elaborated-structure feature, and it will be finally coordinated in global optimization.

B. Sparse Feature Enhancement

Next, to guarantee the performance of denoising and super-resolution of SAR imagery, the ℓ_1 regularizer is introduced to formulate the sparse prior, so that the sparse feature can be quantitatively enhanced. From (25), the coefficient matrix $\mathbf{K}_2$ can be set as $\mathbf{K}_2 = \mathbf{I}$, and the second splitted variable $\mathbf{Z}_2 = \mathbf{X}$. In such cases, the second local optimization can be given as

$$\mathbf{Z}_2^{k+1} = \arg\min_{\mathbf{Z}_2} \ \xi_2\|\text{vec}(\mathbf{Z}_2)\|_1 + \frac{\rho_2}{2}\left\|\mathbf{X}^k - \mathbf{Z}_2 + \mathbf{D}_2^k\right\|_F^2 \tag{33}$$

where $\|\cdot\|_1$ is the ℓ_1-norm operator, and ρ_2 denotes the augmented Lagrangian parameter. According to the proximal algorithm [28], [37], the problem in (33) can be solved in the soft-thresholding operator as

$$\mathbf{Z}_2^{k+1} = \text{prox}_{\frac{\xi_2}{\rho_2}\|\cdot\|_1}\left(\mathbf{X}^k + \mathbf{D}_2^k\right) \tag{34}$$

where $\text{prox}_{(\xi_2/\rho_2)\|\cdot\|_1}(\cdot)$ is the proximal operator of the $\|\cdot\|_1$-norm, which denotes the solution for the sparse feature. Considering the complex value of SAR data, the solution for the sparse variable $\mathbf{Z}_2$ is analytically derived in a complex-value form of soft-thresholding operator [38] as

$$\begin{aligned}\mathbf{Z}_2^k &= \left[\mathbf{W}^k - \frac{\frac{\xi_2}{\rho_2}\mathbf{W}^k}{\left|\mathbf{W}^k\right|}\right]_+ \\ &= \frac{\max\left[\left(\left|\mathbf{W}^k\right| - \frac{\xi_2}{\rho_2}\right), 0\right]}{\max\left[\left(\left|\mathbf{W}^k\right| - \frac{\xi_2}{\rho_2}\right), 0\right] + \frac{\xi_2}{\rho_2}}\left(\mathbf{W}^k\right)\end{aligned} \tag{35}$$

where $\mathbf{W}^k = \mathbf{X}^k + \mathbf{D}_2^k$. Generally, the complex soft-thresholding operator in (35) is a linear operator, which guarantees the phase history of the SAR data, and it is capable of ensuring the performance of cooperation with multiple features' enhancement. Similar to the elaborated-structure feature enhancement, the corresponding dual variable $\mathbf{D}_2$ is updated as

$$\mathbf{D}_2^{k+1} = \mathbf{D}_2^k + \mathbf{X}^k - \mathbf{Z}_2^{k+1}. \tag{36}$$

Obviously, the dual variable measures the residual error between the primal variable $\mathbf{X}$ and the splitted variable $\mathbf{Z}_2$.

C. Coordination of Multiple Features

After the elaborated-structure feature and sparse feature are, respectively, evaluated in closed-form solutions as $\mathbf{Z}_1$ and $\mathbf{Z}_2$, it is necessary to coordinate both the features for the global optima $\mathbf{X}$. Based on the ADMM framework in (25), the coordination can be implemented according to

$$\mathbf{X}^{k+1} = \arg\min_{\mathbf{X}} \frac{1}{2}\|\mathbf{Y} - \mathbf{A}\mathbf{X}\|_F^2 + \frac{\rho}{2}\left\|\mathbf{K}\mathbf{X} - \mathbf{Z}^{k+1} + \mathbf{D}^{k+1}\right\|_F^2 \tag{37}$$

where $\mathbf{K}$ is a gathering of the coefficient matrix of $\mathbf{K_1}$ and $\mathbf{K}_2$. By substituting the specific coefficient matrix, the objective function of (37) can be written as

$$\begin{aligned}\Omega(\mathbf{X}) = \frac{1}{2}\|\mathbf{Y} - \mathbf{A}\mathbf{X}\|_F^2 + \frac{\rho_1}{2}\left\|H\mathbf{X} - \mathbf{Z}_1^{k+1} + \mathbf{D}_1^{k+1}\right\|_\chi^2 \\ + \frac{\rho_2}{2}\left\|\mathbf{X} - \mathbf{Z}_2^{k+1} + \mathbf{D}_2^{k+1}\right\|_F^2.\end{aligned} \tag{38}$$

Obviously, the objective function is inherently a convex problem, and the least-square solution can be easily obtained by setting the derivative of the objective function to be zero. Consequently, the objective solution is derived in a closed-form as

$$\begin{aligned}\mathbf{X}^{k+1} = &\left(\mathbf{A}^{\text{H}}\mathbf{A} + \rho_1 H^* H + \rho_2\mathbf{I}\right)^{-1} \\ &\cdot\left\{\mathbf{A}^{\text{H}}\mathbf{Y} + \left[\rho_1 H^*\mathbf{Z}_1^{k+1} + \rho_2\mathbf{Z}_2^{k+1}\right]\right. \\ &\left. - \left[\rho_1 H^*\mathbf{D}_1^{k+1} + \rho_2\mathbf{D}_2^{k+1}\right]\right\}\end{aligned} \tag{39}$$

where $H^* : \mathbb{C}^{N\times M\times 2\times 2} \to \mathbb{C}^{N\times M}$ denotes the adjoint operator of the Hessian matrix, which can transform the high dimensions of the elaborated-structure feature variable into lower dimensions that is consistent with the primal variable $\mathbf{X}$ [39]. It can be noted from (39) that the primal variable $\mathbf{X}$ is finally updated according to both the splitted variables and dual variables. It means that the coordination for the final image data X merges the feature solutions from local optimizations and the residual errors of individual problems. Both the feature solution and the residual error are coordinated, simultaneously, which will definitely improve the performance of multitask learning. The elaborated-structure is quantified by the proposed E-ST operator and the sparse feature is given in the closed-form soft-thresholding operator. Moreover, both the penalty parameters ρ_1 and ρ_2 are quantitatively formulated, and they can be used to cope with practical applications. Therefore, the proposed ESA-SAR algorithm is capable of enhancing the elaborated-structure and sparse features cooperatively.

V. Processing Procedure

Fig. 3 shows the processing flowchart of the proposed ESA-SAR algorithm, which consists of three main stages of two local optimizations and one global optimization. The proposed algorithm is designed for raw SAR data processing. Prior to the ESA-SAR algorithm, SAR IFP [40] proceeds to correct the systematic range migration. The classic

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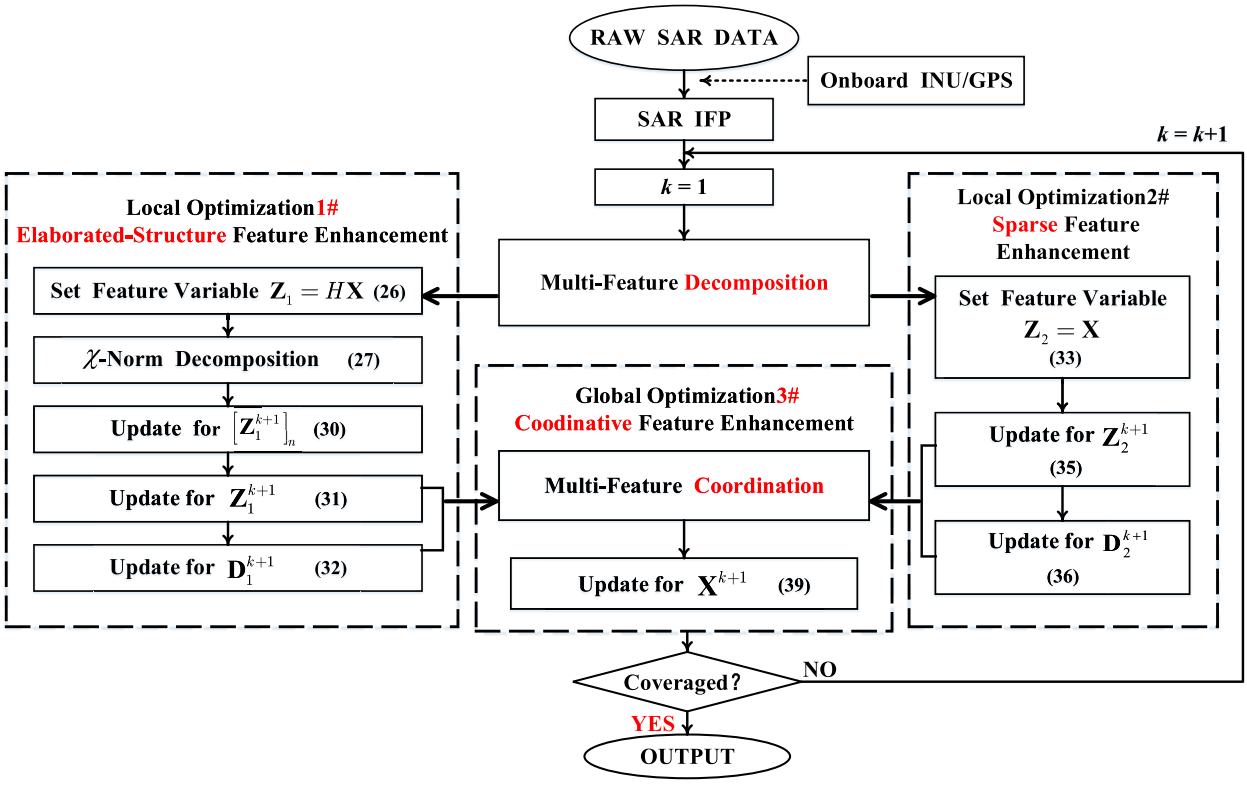


Fig. 3. Processing flowchart of the proposed ESA-SAR algorithm.

range Doppler algorithm (RDA) [41], PFA [42], or back-projection (BP) algorithm [43], etc. can be applied. In the ideal case, the airborne SAR flies accurately along the predetermined flight track. While in nonideal cases, there are motion errors/deviations, and it can be removed by either onboard INU/GPS or data-driven autofocusing [32]. Next, the ESA-SAR algorithm carries out in the "decomposition–coordination" manner according to Section IV. Two splitted variables $\mathbf{Z}_1$ and $\mathbf{Z}_2$ are introduced from the primal variable $\mathbf{X}$ to decompose the joint optimization problem into two local subproblems for the individual feature enhancement.

First, setting $\mathbf{Z}_1 = H\mathbf{X}$ as in (26), the first splitted variable is formulated to capture high-order variations in the clustering feature of the imaging scenario, where the Hessian enhanced TV regularizer is incorporated to encode the prior of elaborated-structure feature. Innovatively, based on the separability of $\|\cdot\|_\chi$-norm as shown in (27), the elaborated-structure feature variable is derived from the independent element in the closed-form E-ST operator as shown in (30). It greatly facilitates the update for the feature variable in an analytical form. It is worth noting that the Hessian regularizer uses the S_p-norm to realize the extraction of eigenvalues which are the main elements of E-ST. Therefore, the Schatten norm not only can realize the eigen information awareness of the elaborated-structure but also facilitate the derivation for the analytic solution of the elaborated clustering feature. The solutions for the feature variable and the dual variable are developed and updated in (31) and (32), respectively.

Second, the sparse prior is encoded, and the second splitted variable $\mathbf{Z}_2 = \mathbf{X}$ is introduced for sparse feature enhancement. The ℓ_1-norm is encoded for the prior of sparse feature, where the proximity operator can be derived as a complex soft-thresholding form. The solution for the feature variable and the dual variable are derived and updated as in (35) and (36), respectively.

Finally, the feature variables and the dual variables are collected and coordinated as the combined feature variable $\mathbf{Z}$ and dual variable $\mathbf{D}$, respectively. Accordingly, the joint objective function with the unique solution is established in (38) and the final SAR image $\mathbf{X}$ can be coordinately updated as shown in (39). Then, the convergence condition can be defined as the residual r_k at iteration k to guarantee the output of the proposed algorithm is an SAR image with enhanced elaborated-structure and sparse features, where the residual can be expressed as

$$r_k = \mathbf{K}\mathbf{X}^{k+1} - \mathbf{Z}^k. \tag{40}$$

In this article, the condition of convergence is set as that the maximum residual r_k is set up to 10^{-5}.

It can be noted from Fig. 3 that the processing procedure works in a "decomposition–coordination" mechanism [26], where the decomposition splits the joint multifeature enhancement into two local features' enhancement, and the coordination merges the distributed features into a global variable for the intended SAR image. Therefore, it ensures the cooperative enhancement of multiple features.

VI. Experiments

In this section, experiments are conducted to verify the performance of the proposed ESA-SAR algorithm. A number

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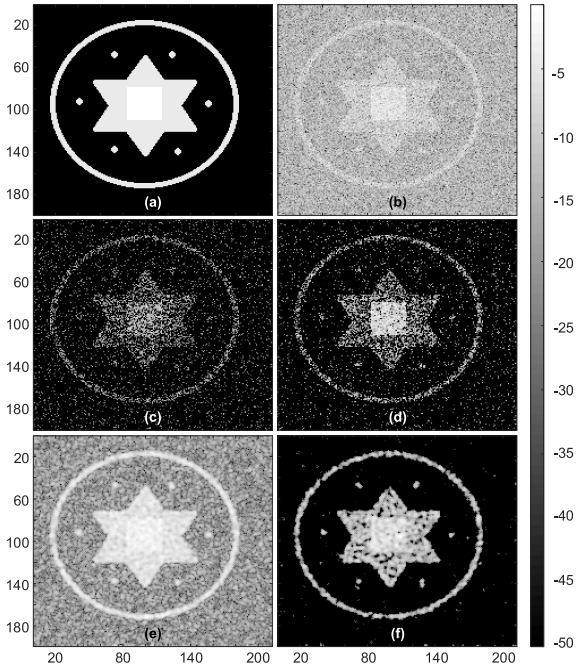


Fig. 4. Simulations. (a) Reference image. (b) Simulated image. (c) Lasso-ADMM. (d) TV-Lasso-ADMM. (e) HETV-ADMM. (f) Proposed ESA-SAR.

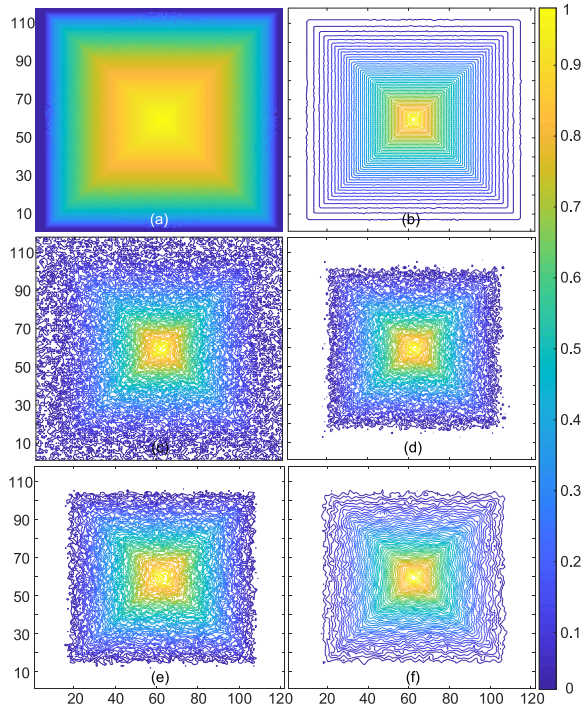


Fig. 5. Contour of the simulations. (a) Simulated reference image. (b) Contour of the simulations. (c) Contour with noise contamination. (d) Lasso-ADMM. (e) TV-Lasso-ADMM. (f) Proposed ESA-SAR.

TABLE I
SIMULATION CORRELATION

Simulated image	0.4690
Lasso-ADMM	0.4422
TV-Lasso-ADMM	0.5549
HETV-ADMM.	0.8797
Proposed ESA-SAR	0.8962

of datasets are provided for validation. Both the simulated and raw SAR data are used to prove the applicability of ESA-SAR algorithm. Comparisons with the conventional algorithms are performed to show the superiority of the proposed algorithm. In addition, the phase transition analysis [44] is used to reveal the performance of the proposed algorithm under various conditions.

A. Simulated Data Experiments

First, the simulated data are generated in the airborne spotlight SAR mode to verify the effectiveness and superiority of the proposed ESA-SAR algorithm. The radar is simulated in the X-band with a centroid frequency of 10 GHz, a bandwidth of 1.2 GHz, an airborne platform height of 6000 m, and a PRF about 1000 Hz. In Fig. 4(a), a simulated imaging scenario is used for simulated data generation, where various complicated structures are simulated, including isolated dominant scatterers, continuous structural regions, and overlapped structures. In the simulations, each scatterer is assumed as isotropic. Totally, 212 pulse echoes with each of 199 range samples are generated. Transforming the data into range compression domain and simulating Gaussian white noise in SNR of -5 dB, the noise-contaminated image can be formed by Fourier transformation into the range-Doppler domain as shown in Fig. 4(b). Fig. 4(c)–(f) are the results by Lasso-ADMM [45] with ℓ_1-norm regularizer, TV-Lasso-ADMM [46], HETV-ADMM, and the proposed ESA-SAR algorithm, respectively, where TV-Lasso-ADMM exploits ℓ_1-norm and TV as regularizer in the ADMM framework. In terms of visual comparison, the proposed ESA-SAR algorithm performs the best to recover the complicated features of interests, where almost the same structures are recovered as the ground truth in Fig. 4(a). Besides, from the comparisons, the Lasso-ADMM only encodes sparse prior, and it is helpless to the recovery of structure feature. TV-Lasso-ADMM is improved only for dominant scatterers. Although the optimization by the HETV-ADMM algorithm can recover the target structure, the strong background noise seriously affects the extraction of detailed structures. Furthermore, because the algorithms in the comparison are all obtained and updated in the ADMM framework, the computational complexity of the algorithms is similar and all in the order of $O(2M \log N)$. By comparing (31) and (34), it can be observed that the E-ST operator of the proposed algorithm is more general than the conventional soft-thresholding operator of sparsity enhancement. In (31), because high-order feature variation is to be captured, the Hessian operator H is incorporated in the proposed algorithm, where the overall computational complexity in elementwise calculation will not increase too much. It has the same order with respect to the soft-thresholding operator [27] in terms of computational complexity. To quantitatively evaluate the performance, the correlation (COR) between the experimental result and the reference image is given in Table I. The

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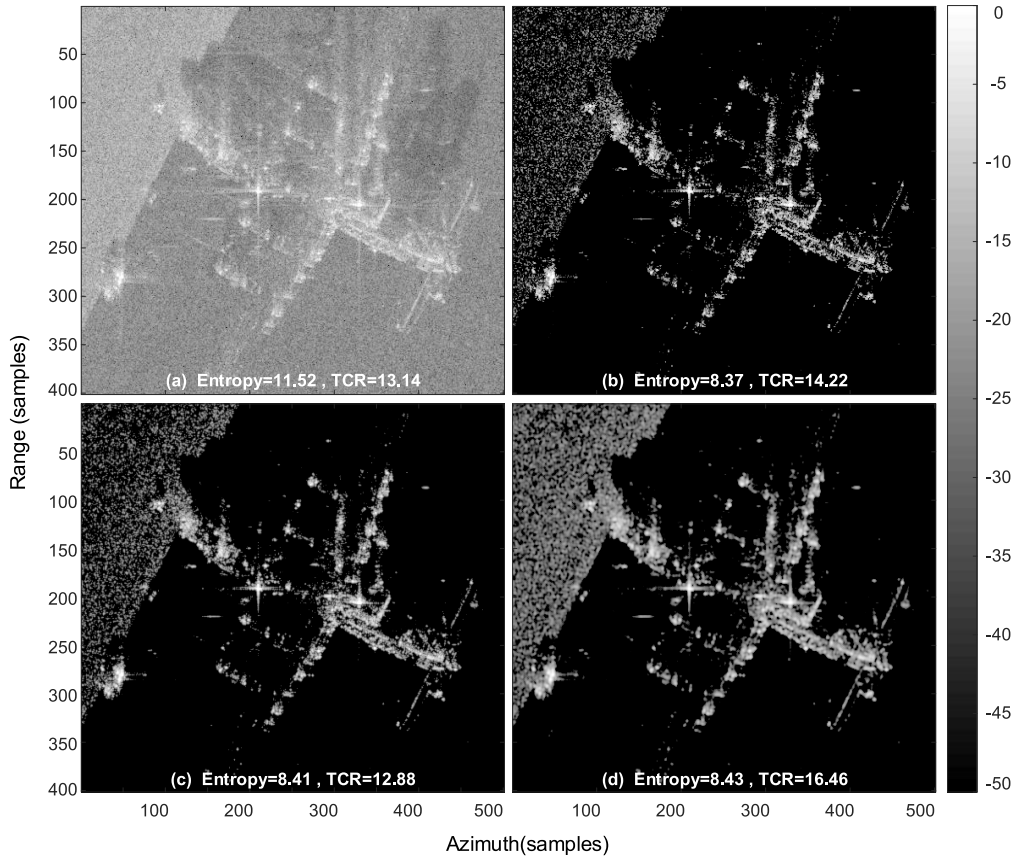


Fig. 6. Comparison of multifeature enhancement of practical SAR dataset (a) without feature enhancement, (b) with sparse feature enhancement, (c) with structural and sparse feature enhancement, and (d) with elaborated-structure and sparse feature enhancement by the proposed ESA-SAR.

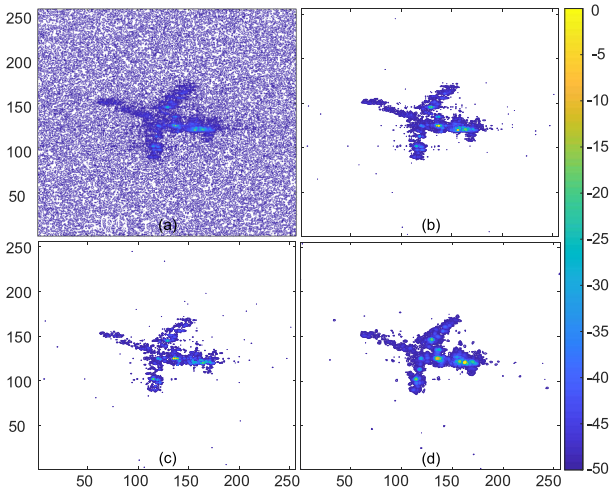


Fig. 7. Results of air-targets' feature enhancement by (a) Fourier transformation under USR 0.8 and SNR 5 dB, (b) Lasso-ADMM, (c) TV-Lasso-ADMM, and (d) ESA-SAR.

correlation is calculated as

$$\mathrm{COR}(\mathbf{X}^k) = \frac{E(\mathbf{X}^k\mathbf{X}) - E(\mathbf{X}^k)E(\mathbf{X})}{\sigma(\mathbf{X}^k)\sigma(\mathbf{X})} \tag{41}$$

where $\mathbf{X}^k$ is the experimental result, $\mathbf{X}$ is the reference image, $E(\cdot)$ denotes the mean, and $\sigma(\cdot)$ denotes the variance. As the quantitative indicator, the larger the correlation, the higher the accuracy of the structure feature enhancement. In terms of quantitative comparison, the proposed ESA-SAR achieves the highest COR, which denotes the proposed algorithm achieves the highest accuracy for the complicated feature recovery and enhancement.

Second, to further examine the performance of the proposed algorithm in complicated imaging scenario, one more simulation is carried out, where a multilayer structure is simulated. As shown in Fig. 5(a), the simulated scenario is set with multiple overlapped flat targets to verify the capability of elaborated-structure awareness, where the darker the area, the more the overlapped structures, and the outer structure is less clear than the inner structure. The simulated data have a dimension of 118 × 122 samples (range × azimuth). For a visual comparison, the contour of imaging results is shown in Fig. 5(b)–(f), where Fig. 5(b) is the contour of Fig. 5(a) and Fig. 5(c) is the result of adding noise to Fig. 5(b). Fig. 5(d) is the result by Lasso-ADMM algorithm for Fig. 5(c), where only the sparse features are enhanced, that is, the noise is suppressed, but the complicated layer structure features are not effectively extracted at all. Fig. 5(e) is the result by TV-Lasso-ADMM algorithm for Fig. 5(c), where compared with Fig. 5(d), the structure of the scene is slightly restored, but not as well as Fig. 5(f), which is the result by the proposed ESA-SAR algorithm. By comparing the results of the three algorithms in Fig. 5(d)–(f), it can be noted that the proposed ESA-SAR algorithm can recover the continuous connection of contours, while other algorithms cannot, where the contours are disconnected and mussy. As shown in the

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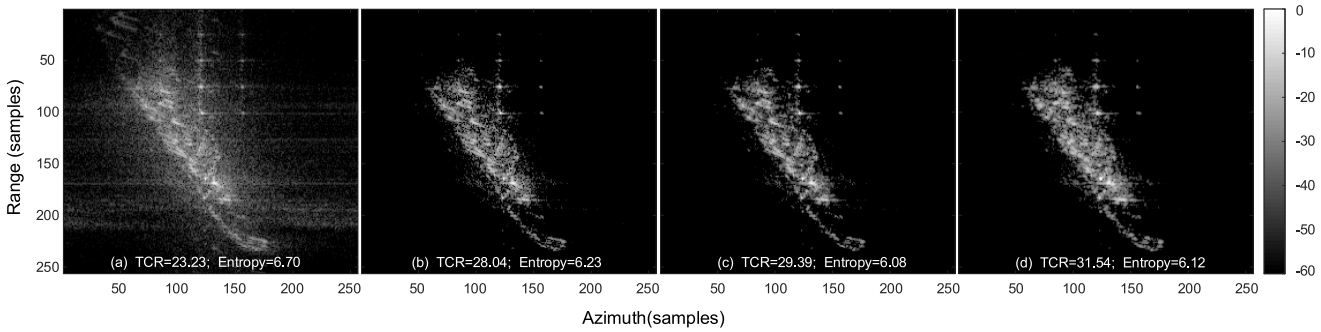


Fig. 8. Results of ISAR ship target. (a) Fourier transformation. (b) Lasso-ADMM. (c) TV-Lasso-ADMM. (d) ESA-SAR.

details, the proposed algorithm achieves the highest accuracy in recovering the layer structures as the ground truth shown in Fig. 5(b). Therefore, it is effective to recover and enhance the elaborated-structure features.

B. Raw SAR Data Experiments

Next, to examine the performance of the proposed ESA-SAR algorithm in practical SAR datasets, the SAR data of ground target, air target, and ship target are used to compare with the conventional algorithms. First, the real SAR image data in high resolution of 0.1 m are processed in terms of feature enhancement, and the results are shown in Fig. 6. The real data were collected in the Ka-band and the spotlight mode. Fig. 6(a) is formed by Fourier transformation and without feature enhancement, where the background clutter is strong and the target of helicopter is with low contrast due to the limited mainlobe width and high sidelobes. In Fig. 6(b), the sparse feature enhancement is performed by the Lasso-ADMM algorithm. Although the background clutter is suppressed well, the features of weak scatterers are not apparent, such as the tail of the plane. To this end, the structural and sparse features are enhanced in a cooperative manner by the TV-Lasso-ADMM algorithm, and the imaging result is shown in Fig. 6(c). Due to the complexity of target structure, the effect of synergetic enhancement for structural and sparse feature enhancement cannot significantly improve the performance over the result of Lasso-ADMM algorithm. In Fig. 6(d), the proposed ESA-SAR algorithm proceeds, and the resultant imaging achieves best performance in terms of contrast, continuity, and integrity of feature interests. To measure the imaging performance quantitatively, the indicators of entropy and target clutter ratio (TCR) are used. The entropy reflects the focusing quality of image. The lower the image entropy, the larger the dynamic range of scattering features, the better the focusing quality. The larger the TCR, the higher the SNR of the image and the more targets can be detected and identified under the relative noise. What is noteworthy is that the TCR of the proposed ESA-SAR algorithm in Fig. 6(d) is the largest, which means that the proposed algorithm can recover the target features in highest SNR. Although the entropy of the proposed ESA-SAR algorithm imaging is not the smallest, it also shows a better contrast than raw data imaging, and the proposed algorithm achieves a good balance between the imaging result of contrast and SNR.

Furthermore, the inverse SAR (ISAR) data are used to verify the validity and superiority of the proposed algorithm. ISAR is theoretically equivalent to SAR under turntable model [26]. The experimental results of the Yak-42 plane are shown in Fig. 7. To verify the robustness of the proposed ESA-SAR algorithm, the noise of 5 dB and USR of 0.8 are simulated in the raw ISAR data, and the image result is shown in Fig. 7(a). The Lasso-ADMM, TV-Lasso-ADMM, and the proposed ESA-SAR are applied, which are shown in Fig. 7(b)–(d), respectively. From the comparison, while the Lasso-ADMM and TV-Lasso-ADMM achieve the sparse enhancement and noise suppression as shown in Fig. 7(b) and (c), the nose position of aircraft is fractured due to the undersampled data. However, the proposed ESA-SAR algorithm is capable of recovering the complete nose and engines of the aircraft, and the background noise is suppressed at the lowest level. The proposed ESA-SAR algorithm is effective to recover and enhance the complicated feature of target of interest.

The other raw SAR dataset is a ship target with rich structural features, such as mast, gunwale, and stern of the ship. The experimental results are shown in Fig. 8, where the Fourier transformation, Lasso-ADMM, TV-ADMM, and the proposed ESA-SAR are used, respectively. A similar conclusion can be observed that the proposed algorithm achieves the best performance in terms of recovering the elaborated-structure features, which can be observed from the recovery of the mast, gunwale, and superstructure. Quantitatively, from the comparison of image entropy and TCR, it can be drawn that the more detailed features of the target can be recovered by the proposed ESA-SAR algorithm. Therefore, the proposed algorithm achieves the highest accuracy for the recovery of complicated and elaborated-structures of target of interest.

C. Performance Evaluation

To quantitatively examine the performance of the proposed algorithm in various noise and reduced sampling conditions, the phase transition analysis is introduced. In Fig. 9, the phase transition diagrams (PTDs) are presented from Lasso-ADMM, TV-ADMM, and the ESA-SAR algorithms, respectively, where the SNR from -4 to 14 dB and USR from 0 to 1 vary to make different conditions. In this experiment, the numerical result is evaluated by a mean value of the correlation metric between the reference image and the results of algorithm in various conditions. To measure the algorithm performance

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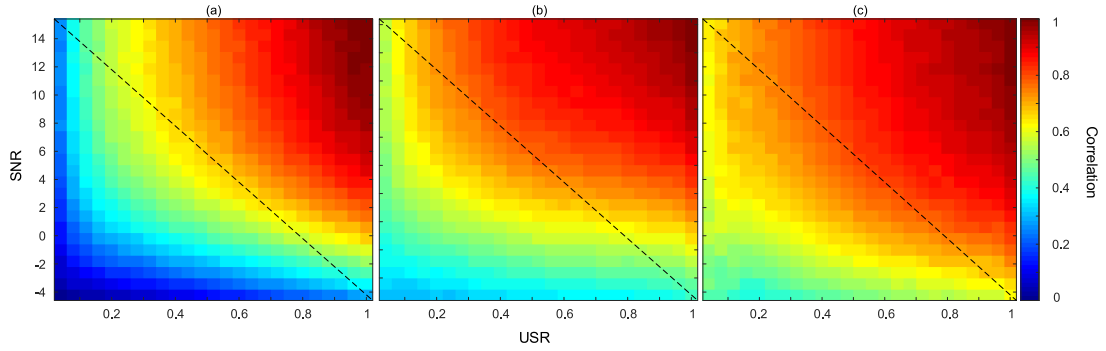


Fig. 9. Phase transition diagram. (a) Lasso-ADMM. (b) TV-Lasso-ADMM. (c) ESA-SAR.

for the elaborated-structure feature recovery, the experimental data are simulated with complicated features for reference. In total, the mean of 100 independent and identical Monte Carlo experiments is provided for each specific SNR and USR conditions. The closer the correlation reaches to 1, the darker the color, and the higher the recovering accuracy of the algorithm. In Fig. 9, the black dotted line is the auxiliary line to compare the area of the dark area. As observed in Fig. 9, the area with large correlation by the proposed ESA-SAR algorithm is the biggest, which means the proposed algorithm is with the best performance in terms of robustness in the elaborated-structure feature recovery.

VII. Conclusion

In this article, a novel ESA-SAR algorithm is proposed to achieve the recovery and enhancement of elaborated-structure feature of target of interest. Specifically, the proposed Hessien-enhanced TV regularizer is designed and incorporated in the ESA-SAR algorithm, which imposes a Hessian operator to access high-order feature information. More specifically, the HETV regularizer projects the Hessian matrix onto the Schatten norm balls, where the high-order structure tensor is established and the elaborated-structure features can be extracted under the fitting of variational directions. Moreover, based on the Schatten norm, the proposed regularizer is analytically solved under the proximal algorithm and derived as an E-ST operator. For the proposed ESA-SAR algorithm, the HETV regularizer and ℓ_1-norm are cooperatively optimized within the ADMM framework and worked in a "decomposition–coordination" procedure. Finally, both the simulated and raw SAR data are used to verify the effectiveness and superiority of the proposed ESA-SAR algorithm, and the phase transition analysis is carried out to demonstrate robustness, quantitatively.

Appendix A

A. Adjoint of the Discrete Gradient Operator

To verify the adjoint relationship between the operators of gradient and divergence as in Section III-A, we first set variable $\mathbf{u} \in \mathbb{C}$ and its dual variable $\mathbf{v} \in \mathrm{sym}^2(\mathbb{C}^2)$ as

$$\mathbf{v}_n = \begin{bmatrix} v_{11} & v_{12} \\ v_{21} & v_{22} \end{bmatrix} \tag{42}$$

where $v_{12} = v_{21}$. According to the rules of Euclid space, for the variable $\mathbf{u}_n$, the inner product of its gradient $\nabla \mathbf{u}_n$ and dual variable is calculated as

$$\begin{aligned} &\sum_n \langle \nabla \mathbf{u}_n, \mathbf{v}_n \rangle_2 \\ &\quad = \sum_n [\nabla \mathbf{u}_n]^T \mathbf{v}_n \\ &\quad = \sum_n ([\nabla_1 \mathbf{u}_n] v_{11} + [\nabla_2 \mathbf{u}_n] v_{21}, [\nabla_1 \mathbf{u}_n] v_{12} + [\nabla_2 \mathbf{u}_n] v_{22}) \end{aligned} \tag{43}$$

where ∇_1 and ∇_2 denote the gradient operators along the azimuthal and slant-range dimension, respectively. Combined with the definition of adjoint operator $\langle Ax, y \rangle_{T_1} = \langle x, A^* y \rangle_{T_2}$, (43) can be derived as

$$\begin{aligned} \sum_n \langle \nabla \mathbf{u}_n, \mathbf{v}_n \rangle_2 &= \sum_n \mathbf{u}_n \left[\nabla_1^* v_{11} + \nabla_2^* v_{21}, \nabla_1^* v_{12} + \nabla_2^* v_{22} \right] \\ &= \sum_n [\mathbf{u}_n] \left[\nabla^* \mathbf{v}_n \right]^T \end{aligned} \tag{44}$$

where ∇_1^* and ∇_2^* denote the adjoint operators of gradients ∇_1 and ∇_2, respectively. In (44), the adjoint operators of gradient are derived as

$$\nabla^* \mathbf{v}_n = \left[\nabla_1^* v_{11} + \nabla_2^* v_{21}, \nabla_1^* v_{12} + \nabla_2^* v_{22} \right]^T. \tag{45}$$

Furthermore, there is the discrete first-order divergence, which is denoted as

$$\mathrm{div} \mathbf{v}_n = \left[\nabla_1^* v_{11} + \nabla_2^* v_{21}, \nabla_1^* v_{12} + \nabla_2^* v_{22} \right]^T. \tag{46}$$

It can be noted that (45) is equivalent to (46), and the equivalence between the gradient and divergence operator can be obtained as

$$\nabla^* \mathbf{v} = \mathrm{div} \mathbf{v}. \tag{47}$$

Consequently, the adjoint relationship between the first-order gradient and divergence can be proved.

B. Adjoint of the Discrete Hessian Operator

To prove the adjoint relationship between the Hessian and second-order divergence as shown in (15) and (16), the

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definition of χ-norm is first given as

$$\sum_n \langle H\mathbf{u}, \mathbf{v}\rangle_\chi = \sum_n \mathrm{tr}\langle H\mathbf{u}_n, \mathbf{v}_n\rangle_2 \tag{48}$$

where $\mathrm{tr}(\cdot)$ denotes the trace of a matrix. According to the definition of χ-norm, for the variable $\mathbf{u}_n$, the inner product of its Hessian matrix H_n and dual variable is calculated as

$$\begin{aligned}
&\sum_n \mathrm{tr}\langle H\mathbf{u}_n, \mathbf{v}_n\rangle_2 \\
&\quad= \sum_n \mathrm{tr}\big([H\mathbf{u}_n]^T\mathbf{v}_n\big) \\
&\quad= \sum_n \big(\,\big[\nabla_{11}^2\mathbf{u}\big]v_{11} + 2\big[\nabla_{12}^2\mathbf{u}\big]v_{21} + \big[\nabla_{22}^2\mathbf{u}\big]v_{22}\,\big) \\
&\quad= \sum_n \mathbf{u}_n\big(\,\big[\nabla_{11}^{*2}\big]v_{11} + 2\big[\nabla_{12}^{*2}\big]v_{21} + \big[\nabla_{22}^{*2}\big]v_{22}\,\big)
\end{aligned} \tag{49}$$

where ∇_{11}^2, ∇_{22}^2, and ∇_{12}^2 denote the second-order gradient operators, and ∇_{ij}^{*2} denotes their associated adjoint operators. In (50), the adjoint operators of discrete Hessian operator is derived as

$$H^*(\mathbf{v}_n) = \big[\nabla_{11}^{*2}\big]v_{11} + 2\big[\nabla_{12}^{*2}\big]v_{21} + \big[\nabla_{22}^{*2}\big]v_{22}. \tag{50}$$

Furthermore, based on the first-order divergence in (46), the second-order discrete divergence can be denoted as

$$\mathrm{div}^2\mathbf{v}_n = \big[\nabla_{11}^{*2}\big]v_{11} + 2\big[\nabla_{12}^{*2}\big]v_{21} + \big[\nabla_{22}^{*2}\big]v_{22}. \tag{51}$$

It can be noted that (50) is equal to (51), and there is

$$H^*(\mathbf{v}_n) = \mathrm{div}^2\mathbf{v}_n. \tag{52}$$

Consequently, the adjoint relationship between the Hessian operator and the second-order divergence is proven.

Appendix B

Based on the proof of the adjoint relationship between gradient and divergence, the equivalent relationship between different expressions for TV and HETV regularizer can be derived in detail.

A. Proof of Equivalent Relationship for TV Regularizer

To verify the equivalent relationship between (9) and (10), the definition of the dual norm is first introduced. Assuming dual norm of the norm $\|z\|$ is denoted as $\|z\|_*$, which has specific expression as

$$\|z\|_* = \sup\big\{z^T x, \|x\| \le 1\big\}. \tag{53}$$

Based on the definition of dual norm, the TV regularizer shown in (10) can be deduced as

$$\mathrm{TV}(\mathbf{u}) = \sum_n \|\nabla\mathbf{u}\|_2 = \sum_n \sup_{\mathbf{v}_n\in B_2} \langle\nabla\mathbf{u}_n, \mathbf{v}_n\rangle_2 \tag{54}$$

where $B_2 = \{\mathbf{v}_n, \|\mathbf{v}_n\|_2 \le 1\}$ denotes a unit Euclidean norm sphere. According to the definition of adjoint operator, the TV regularizer can be further expressed as

$$\begin{aligned}
\mathrm{TV}(\mathbf{u}) &= \sum_n \sup_{\mathbf{v}_n\in B_2} \langle\nabla\mathbf{u}_n, \mathbf{v}_n\rangle_2 = \sum_n \sup_{\mathbf{v}_n\in B_2} \big\langle\mathbf{u}_n, \nabla^*\mathbf{v}_n\big\rangle_2 \\
&= \sum_n \sup_{\mathbf{v}_n\in B_2} \langle\mathbf{u}_n, \mathrm{div}(\mathbf{v}_n)\rangle_2.
\end{aligned} \tag{55}$$

By transforming the discrete form of (55) into continuous form, (9) of TV regularizer can be derived as

$$\begin{aligned}
\mathrm{TV}(\mathbf{u}) = \sup\Big\{ \int_{\boldsymbol{\Omega}} \mathbf{u}(x)(\mathrm{div}\mathbf{v}(x))dx \mid \mathbf{v}\in\mathbf{C}_c^1(\boldsymbol{\Omega}, \mathbf{C}^2), \\
\mathbf{v}(x)\in B_2(0), \forall x\in\boldsymbol{\Omega}\Big\}.
\end{aligned} \tag{56}$$

Therefore, the equivalence of (9) and (10) can be proved.

B. Proof of Equivalent Relationship for HETV Regularizer

In this section, the equivalent relationship between (14) and (17) is elaborately derived. Different from TV regularizer, the HETV regularizer projects the second-order structure tensor onto the Schatten norm balls instead of the Euclidean norm balls. Based on the theoretical basis provided above, (14) can be derived as

$$\begin{aligned}
&\mathrm{HETV}(\mathbf{u}) \\
&\quad= \sup\Big\{ \int_{\boldsymbol{\Omega}} \mathbf{u}(x)\big(\mathrm{div}^2\mathbf{v}(x)\big)dx \mid \mathbf{v}\in\mathbf{C}_c^1(\boldsymbol{\Omega}, \mathbf{C}^2), \\
&\qquad\qquad \mathbf{v}(x)\in B_{Sq\times Sq}(0), (\mathrm{div}\mathbf{v}(x))^T \in B_{Sq}(0), \forall x\in\boldsymbol{\Omega}\Big\} \\
&\quad= \sum_n \sup_{\mathbf{v}_n\in B_{Sq\times Sq}(0)} \big\langle\mathrm{div}^2\mathbf{v}_n, \mathbf{u}_n\big\rangle_2 \\
&\quad= \sum_n \sup_{\mathbf{v}_n\in B_{Sq\times Sq}(0)} \langle H\mathbf{v}_n, \mathbf{u}_n\rangle_2
\end{aligned} \tag{57}$$

where $B_{Sq} = \{\mathbf{v}_n, \|\mathbf{v}_n\|_{Sq} \le 1\}$ denotes a q-order Schatten norm sphere. According to the definition of dual norm, Sp-norm is the dual norm of Sq-norm, which satisfies $(1/q) + (1/p) = 1$. Based on the definition of adjoint operator, HETV regularizer in (57) can be derived as

$$\mathrm{HETV}(\mathbf{u}) = \sum_n \sup_{\mathbf{v}_n\in B_{Sq\times Sq}(0)} [H\mathbf{u}_n]^T\mathbf{v}_n. \tag{58}$$

According to the operation rules of the dual norm, the discrete expression of HETV regularizer can be derived as

$$\mathrm{HETV}(\mathbf{u}) = \sum_n \|H\mathbf{u}_n\|_{S_p}. \tag{59}$$

Therefore, the equivalence of (14) and (17) can be proved.

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